

Carlos Castillo-Passi

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Education

- 2025–Present **Postdoctoral Scholar**, *Stanford University*, California, USA
- 2022–2024 **Joint PhD in Biological and Medical Eng.**, *King's College London*, London, UK
- 2019–2024 **PhD in Biological and Medical Engineering (Ranked #1)**, *PUC*, Santiago, Chile
- 2015–2018 **MSc in Electrical Engineering (Highest Distinction)**, *PUC*, Santiago, Chile
- 2011–2018 **Electrical Engineering**, *PUC*, Santiago, Chile

Academic experience

Teaching

- 2026 **Primary Instructor**, BIOS214 Open-Source MRI Sequence Development: Simulation to Scanner (Autumn 2026), *Stanford University*, California, USA
- 2025 **Guest Lecturer**, RAD229 MRI Sequences and Signals (Spring 2025), *Stanford University*, California, USA
- 2024 **2-Day Workshop - Advanced Scientific Computing using Julia** - librehub.github.io/2024/12/19/julia-ws-2024.html, *PUC*, Santiago, Chile
- 2019 **Diploma in Fundamentals and Applications of Magnetic Resonance**, *PUC*, Santiago, Chile
- 2014–2015 **Developed 4 Precalculus MOOCs** – online.ing.uc.cl, *PUC*, Santiago, Chile

Teaching assistant

Electrical Engineering (PUC):

- IEE2103 Signals and Systems [2014–2020]
- IEE2513 Communications
- IEE3764 Image Formation
- IEE3794 Image Reconstruction [2019–2021]

Medical Imaging Sciences (KCL):

- 7MIGEP43 Medical Imaging Acquisition
- 7MIGEP45 Medical Image Reconstruction

Biomedical Engineering (KCL):

- 5CCYB050 Signal and Image Processing

Mathematical Engineering (PUC):

- IMT3840 Topics in Inverse Problems

Biomedical Engineering (PUC):

- IBM2101 Biomedical Imaging
- IBM2122 Biomedical Design I [2016–2018]
- IBM2122 Biomedical Design II [2016–2018]
- IBM3103 Mathematical Methods for BME
- IBM3102 Advanced Topics in BME II

Selected scientific publications

- 2026 **Versatile and Highly Efficient MRI Simulation of Arbitrary Motion in KomamMRI**, Pablo Villacorta-Aylagas, **Carlos Castillo-Passi**, Ryan Anders Kierulf, Rosa María Menchón-Lara, Justino R. Rodríguez-Galván, José Benito Sierra-Pallares, Pablo Irarrazaval, and Carlos Alberola-López. *Magnetic Resonance in Medicine*. Paper.
- 2026 **Bringing Julia to Health: A Concise Overview of JuliaHealth**, Jacob S. Zelko, **Carlos Castillo-Passi**, Jakub Mitura, Divyansh Goyal, Kosuri Lakshmi Indu, and Jay Sanjay Landge. *Proceedings of the JuliaCon Conferences*. Paper.

- 2025 **Simultaneous liver T1, T2, and ADC MR fingerprinting using optimized motion-compensated diffusion preparations: An initial validation on volunteers**, Carlos Velasco, **Carlos Castillo-Passi**, Nadia Chaher, Dimitrios Karampinos, Pablo Irarrazaval, Alkystis Phinikaridou, René Botnar, and Claudia Prieto. *Magnetic Resonance in Medicine*. Carlos Velasco and **Carlos Castillo-Passi** contributed equally. Paper.
- 2025 **Simultaneous 3D aortic lumen and vessel wall imaging at 0.55 T at either systole or diastole**, Matias Paredes, **Carlos Castillo-Passi**, Karl P. Kunze, Anastasia Fotaki, Simon Littlewood, René M. Botnar, and Claudia Prieto. *Magnetic Resonance in Medicine*. Paper.
- 2025 **Highly efficient image navigator based 3D whole-heart cardiac MRA at 0.55T**, **Carlos Castillo-Passi**, Karl P. Kunze, Michael G. Crabb, Camila Munoz, Anastasia Fotaki, Radhouene Neji, Pablo Irarrazaval, Claudia Prieto, and René M. Botnar. *Magnetic Resonance in Medicine*. Paper.
- 2024 **Fast and Accessible T2 Mapping Using Off-Resonance Corrected DESPOT2 With Application to 3D Prostate**, Ronal Coronado, **Carlos Castillo-Passi**, Cecilia Besa, and Pablo Irarrazaval. *Magnetic Resonance Imaging*. Paper.
- 2023  **KomaMRI.jl: An Open-Source Framework for General MRI Simulations with GPU Acceleration**, **Carlos Castillo-Passi**, Ronal Coronado, Gabriel Varela-Mattatall, Carlos Alberola-López, René Botnar, and Pablo Irarrazaval. *Magnetic Resonance in Medicine*. Paper.
- 2022 **Accurate characterization of dynamic microbial gene expression and growth rate profiles**, Gonzalo Vidal, Carolus Vitalis, Macarena Muñoz Silva, **Carlos Castillo-Passi**, Guillermo Yáñez Feliú, Fernán Federici, and Timothy J. Rudge. *Synthetic Biology*. Paper.
- 2022 **DeepSPIO: Super Paramagnetic Iron Oxide Particle Quantification using Deep Learning in Magnetic Resonance Imaging**, Gabriel Della Maggiora, **Carlos Castillo-Passi**, Wenqi Qiu, Shuang Liu, Carlos Milovic, Masaki Sekino, Cristian Tejos, Sergio Uribe, and Pablo Irarrazaval. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. Paper.
- 2021 **A spatial off-resonance correction in spirals for Magnetic Resonance Fingerprinting**, Ronal Coronado, Gastao Cruz, **Carlos Castillo-Passi**, Cristian Tejos, Sergio Uribe, Claudia Prieto, and Pablo Irarrazaval. *IEEE Transactions on Medical Imaging*. Paper.
- 2020 **MAPL1: q-space reconstruction using ℓ_1 -regularized mean apparent propagator**, Gabriel Varela-Mattatall, **Carlos Castillo-Passi**, Alexandra Koch, Joaquin Mura, Rüdiger Stirnberg, Sergio Uribe, Cristian Tejos, Tony Stöcker, and Pablo Irarrazaval. *Magnetic Resonance in Medicine*. Paper.

International conference participation

- 2026 **[Invited educational talk] Introduction to numerical phantoms and MRI sequence simulation with KomaMRI**, **Carlos Castillo-Passi** and Pablo Villacorta. *MRITogether 2026 · Virtual meeting*.
- 2026 **[Invited educational talk] Open-Source Frameworks for MRI Reconstruction II**, **Carlos Castillo-Passi**. *Demystifying MRI Reconstruction: Classical Foundations to AI Frontiers, ISMRM 2026 · Cape Town, South Africa*.
- 2026 **[Conference talk] How I Drew the Julia Logo Using Spins in an MRI Machine**, **Carlos Castillo-Passi**. *Health Mini Symposium, JuliaCon 2026 · Mainz, Germany*. Abstract.
- 2026 **[Conference talk] State of JuliaHealth**, Hetarth Shah, **Carlos Castillo-Passi**, and Jacob Zelko. *Health Mini Symposium, JuliaCon 2026 · Mainz, Germany*. Abstract.
- 2026 **[Digital poster] Highly-Efficient RF Pulse Design via Compiler-Level Reverse-Mode Automatic Differentiation of GPU-Accelerated MRI Simulations**, Kareem Fareed, Charles McGrath, Jakub Mitura, Daniel B Ennis, and **Carlos A Castillo-Passi**. *ISMRM 2026 · Cape Town, South Africa*. Presented by Kareem Fareed Abstract.

- 2026 **[Power pitch] Simulation and Evaluation of Joint Velocity-Acceleration Encoded 4D-Flow MRI**, Charles McGrath, **Carlos Castillo-Passi**, Pablo Villacorta-Aylagas, Michael Loecher, and Daniel B. Ennis. *ISMRM 2026 · Cape Town, South Africa*. Presented by Charles McGrath Abstract.
- 2026 **[Power pitch] Fast and accurate Bloch simulations using Magnus expansions**, **Carlos Castillo-Passi**, Charles McGrath, Kareem Fareed, Jacob S. Blum, and Daniel B. Ennis. *ISMRM 2026 · Cape Town, South Africa*. Abstract.
- 2026 **[Oral presentation] Fast and accurate Bloch simulations using Magnus expansions**, **Carlos Castillo-Passi**, Charles McGrath, Kareem Fareed, Jacob S. Blum, and Daniel B. Ennis. *ISMRM 2026 Workshop on Data Sampling and Image Reconstruction · Sedona, USA*. Abstract.
- 2025 **[Invited talk] Go with the Flow: Using Extensible and Shareable Motion Phantoms in KomaMRI**, **Carlos Castillo-Passi**. *MRITogether 2025 · Virtual meeting*.
- 2025 **[Invited educational talk] Modern Open-Source MRI Simulations**, **Carlos Castillo-Passi**. *Open Innovation in MR from Vendor & Academia Perspective, ISMRM 2025 · Honolulu, USA*.
- 2025 **[Conference talk] What's New with KomaMRI.jl**, **Carlos Castillo-Passi**. *JuliaHealth Minisymposium, JuliaCon 2025 · Pittsburgh, USA*. Abstract.
- 2025 **[Oral presentation] KomaMRI.jl Device-Agnostic, Highly Efficient, and Pulseseq-compatible MRI Simulations**, **Carlos Castillo-Passi**, Pablo Villacorta, Justino Rodriguez-Galvan, Carlos Alberola, Claudia Prieto, Pablo Irarrazaval, and René M. Botnar. *SCMR 2025 · Washington, DC, USA*. Abstract.
- 2024 **[Invited educational talk] Interactive Visualizations**, **Carlos Castillo-Passi**. *Unlocking Productivity & Impact in Teaching & Publishing II, ISMRM 2024 · Singapore*.
- 2024 **[Oral presentation] Liver T1, T2 and ADC Magnetic Resonance Fingerprinting in a single breath-hold at 3T**, Carlos Velasco, **Carlos Castillo-Passi**, Nadia Chaher, Alkystis Phinikaridou, René M. Botnar, and Claudia Prieto. *ISMRM 2024 · Singapore*. Abstract.
-  2024 **[Power pitch] Cardiac Magnetic Resonance Fingerprinting for Simultaneous T1 and T2 Mapping at 0.55T**, **Carlos Castillo-Passi**, Carlos Velasco, Donovan Tripp, Karl P. Kunze, Radhouene Neji, Pablo Irarrazaval, René M. Botnar, and Claudia Prieto. *ISMRM 2024 · Singapore*. Abstract.
- 2024 **[Abstract] Magnetic Resonance Fingerprinting for Simultaneous T1, T2 and ADC Mapping at 0.55T using Convex-optimized Diffusion Prepared Waveforms**, **Carlos Castillo-Passi**, Carlos Velasco, Donovan Tripp, Karl P. Kunze, Radhouene Neji, Pablo Irarrazaval, René M. Botnar, and Claudia Prieto. *ISMRM 2024 · Singapore*. Abstract.
- 2024 **[Oral presentation] Simultaneous iNAV-based 3D Whole-Heart Bright-Blood and Black-Blood Imaging at 0.55T**, **Carlos Castillo-Passi**, Michael Crabb, Camila Munoz, Donovan Tripp, Karl Kunze, Radhouene Neji, Pablo Irarrazaval, Claudia Prieto, and René Botnar. *SCMR 2024 · London, United Kingdom*. Abstract.
-  2024 **[Abstract] Cardiac Magnetic Resonance Fingerprinting for Simultaneous T1 and T2 Mapping at 0.55T**, **Carlos Castillo-Passi**, Carlos Velasco, Karl Kunze, Radhouene Neji, Pablo Irarrazaval, René Botnar, and Claudia Prieto. *SCMR 2024 · London, United Kingdom*. Abstract.
- 2023 **[Invited talk] Using KomaMRI.jl for Comprehensive Quantitative MRI**, **Carlos Castillo-Passi**. *Vendor-Agnostic Pulse Sequence Programming with Pulseq: From Basics to Advanced Topics, ISMRM 2023 · Virtual meeting*.
- 2023 **[Invited talk] KomaMRI.jl: Framework for MRI Simulations with GPU Acceleration**, **Carlos Castillo-Passi**. *MRITogether 2023 · Virtual meeting*.
- 2023 **[Conference talk] KomaMRI.jl: Framework for MRI Simulations with GPU Acceleration**, **Carlos Castillo-Passi**. *JuliaCon 2023 · MIT, Cambridge, USA*. Abstract.

2023  [Oral presentation] Whole-Heart Cardiovascular Magnetic Resonance Angiography with iNAV-based Non-Rigid Motion-Corrected Reconstruction at 0.55T, Carlos Castillo-Passi, Michael G. Crabb, Camila Muñoz, Karl P. Kunze, Radhouene Neji, Pablo Irarrazaval, Claudia Prieto, and René M. Botnar. *ISMRM 2023 · Toronto, Canada*. Abstract.

2023 [Oral presentation] Whole-Heart Cardiovascular Magnetic Resonance Angiography with iNAV-based Non-Rigid Motion-Corrected Reconstruction at 0.55T, Carlos Castillo-Passi, Michael G. Crabb, Camila Munoz, Karl P. Kunze, Radhouene Neji, Pablo Irarrazaval, Claudia Prieto, and René M. Botnar. *SCMR 2023 · San Diego, USA*. Abstract.

2022 [Abstract] MRIsim.jl: A framework for end-to-end spin-level MRI simulations with GPU acceleration, Carlos Castillo-Passi, Ronal Coronado, Sergio Uribe, and Pablo Irarrazaval. *ISMRM & SMRT Annual Meeting 2022 · London, United Kingdom*. Abstract.

2018–2021 And 9 other abstracts

Book chapters

2026 Cardiac MRI: Myocardial tissue characterization, D. Ennis, C. Castillo-Passi, S. O. A. Thalen, and A. Hannum. *Biomechanics of the Heart: From Patient to Model and Back*, Elsevier. Book chapter · In press.

Grants/Fellowships

2026 NumFOCUS Small Development Grant (Julia Health, \$10,000), NumFOCUS

Achievements and awards

2026 3rd Place Trainee Abstract Award: “Fast and Accurate Bloch Simulations using Magnus Expansions”, *ISMRM Open & Reproducible Research Study Group*

2026 Top Viewed Article in MRM (Top 10%): “Highly efficient image navigator based 3D whole-heart cardiac MRA at 0.55T”, Wiley

2025 APS DCOMP Travel Award (JuliaCon 2025), American Physical Society, Division of Computational Physics (DCOMP)

2025 Top Viewed Article in MRM (Top 10%): “KomaMRI.jl: An Open-Source Framework for General MRI Simulations with GPU Acceleration”, Wiley

2024 2nd Place for Liver T1, T2, and ADC at 3T, Trainee Abstract Presentations, ISMRM MR in Drug Research Business Meeting

2024 Magna Cum Laude Merit Award (Top 15%) for Liver T1, T2, and ADC at 3T, *ISMRM Annual Meeting*, Singapore, Singapore

2024 Early Career Award (1st Place) - Basic Science for iT2prep-BOOST at 0.55T, *CMR2024*, London, United Kingdom

2023 Summa Cum Laude Merit Award (Top 5%) and part of MRM’s fast-track initiative for CMRA at 0.55T, *ISMRM Annual Meeting*, Toronto, Canada

2023 Editor’s pick July 2023 for KomaMRI.jl, *Magnetic Resonance in Medicine (MRM)*

2020 Recognition for his outstanding work as a teacher assistant, *Institute of Biological and Medical Engineering*, PUC, Santiago, Chile

2018 Recognition for the making of the MOOCs: EINP001, EINP002, EINP003 and EINP004, PUC, Santiago, Chile

Mentoring

- 2026 **Kosuri Indu - Improving JuliaHealth Documentation Accessibility for Community Onboarding**, *NumFOCUS*, Small Development Grant (SDG), USA
- 2026 **Alya Kurt**, *Summer Undergraduate Research Fellowship (SURF)*, Stanford University, USA
- 2026 **Anvika Singhal - komamripy Python Wrapper for KomaMRI.jl**, *Julia*, Google Summer of Code (GSoC), USA
- 2026 **Kareem Fareed - Differentiable GPU-accelerated MRI Simulations using Reactant.jl**, *Julia*, Google Summer of Code (GSoC), USA
- 2026 **Kareem Fareed - Enabling Fast, Arbitrary, MRI RF Pulse Design**, *Research Experience for Undergraduates (REU)*, Stanford University, USA
- 2025 **Kareem Fareed - Compiler-Level Reverse-Mode Automatic Differentiation (AD) of GPU-Accelerated MRI Simulations**, *Research Experience for Undergraduates (REU)*, Stanford University, USA
- 2024 **Ryan Kierulf - Enhancements to KomaMRI.jl GPU Support**, *Julia*, Google Summer of Code (GSoC)